

Mercure gazeux.

1 1) $C_v = \left. \frac{\partial U}{\partial T} \right|_v$

1 2) $C_p = \left. \frac{\partial H}{\partial T} \right|_p$

1 3) $C_p - C_v = nR$ pour 1 GP.

1 4) $\gamma = \frac{C_p}{C_v}$

1 5) Mercure : GPM $u = \frac{3}{2} nRT$

1 6) $C_v = \frac{3}{2} nR$

1 7) $C_p = \frac{5}{2} nR$

1 8) $\gamma = \frac{C_p}{C_v} = \frac{5}{3}$

9)

2 (a) Eq. état des GP.

$$V_A = \frac{nRT_A}{P_A}$$

$$V_B = \frac{nRT_A}{3P_A} = \frac{V_A}{3}$$

2 (b) $\delta W = -P_{ext} dV$
 $W = - \int_{V_A}^{V_B} P_{ext} dV$

$V_B < V_A$ donc $W > 0$
 $P_{ext} > 0$

2 (c) $\Delta U = C_v \Delta T = W + Q$ (1^{er} principe)

$T_B = T_A \Rightarrow \Delta U = 0 \Rightarrow Q = -W$ donc $Q < 0$

la transf. ne peut pas être adiabatique.

(d) . (i) $W = \int -P_{ext} dV = - \int P dV$ car méca^{ng} réversible.

2 c'est l'oppo^s de l'aire orientée sous la courbe.

$$W = -A$$

Donc les 3 cas $W > 0$.

On a $|A_2| > |A_3| > |A_1|$ donc $W_2 > W_3 > W_1$

(ii) Calcul de W

• isotherme: $PV = nRT$ $P = \frac{nRT}{V}$

$$\begin{aligned} W_1 &= - \int P \cdot dV = - nRT \int \frac{dV}{V} \\ &= - nRT \ln \frac{V_F}{V_i} \\ &= - nRT \ln \frac{V_B}{V_A} \end{aligned}$$

$$\frac{V_B}{V_A} = \frac{1}{3}$$

2 $W_1 = nRT_A \ln 3$

• isochore et isobare

$$\begin{aligned} \underline{W_2} &= W_{A \rightarrow C} + W_{C \rightarrow B} = 0 + W_{C \rightarrow B} \\ &= - P_B \int dV \end{aligned}$$

$$P_B V_B = P_A V_A$$

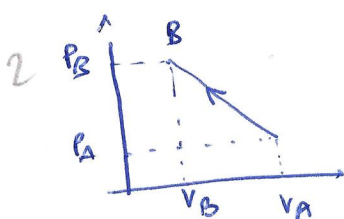
$$\text{car } T_A = T_B$$

$$P_B = P_A \frac{V_A}{V_B}$$

$$= 3P_A$$

2
$$\begin{aligned} &= - \frac{P_B (V_B - V_A)}{1} \\ &= - P_B V_B + P_B V_A \\ &= - nRT_B + 3P_A V_A \\ &= - nRT_B + 3nRT_A \\ &= \underline{2nRT_A} \end{aligned}$$

• San nom



$$|A| = P_A (V_A - V_B) + \frac{1}{2} (P_B - P_A) \cdot (V_A - V_B)$$

Après simplification

$$W_3 = \frac{4}{3} nRT_A$$

$$W_3 > 0 \quad (\text{récepteur})$$

10. (a) $P_1 V_1 = n R T_1$

$$n = \frac{P_1 V_1}{R T_1} = \frac{0,3 \cdot 10^5 \cdot 1}{R \times 573} = \underline{6,3 \text{ mol.}}$$

2

$$P_2 V_2 = n R T_2 \quad V_2 = V_1$$

$$P_2 = \frac{n R T_2}{V_1} = \frac{P_1 \cancel{V_1}}{\cancel{R T_1}} \cdot \frac{R T_2}{\cancel{V_1}} = \underline{P_1 \frac{T_2}{T_1}}$$

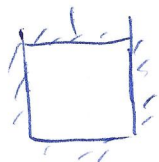
(b). $W = 0$ car transf. isotherme.

$$\begin{aligned} Q &= \Delta U = C_V \Delta T \\ &= C_V (T_2 - T_1) \\ &= C_V \left(\frac{P_2 V_1}{nR} - T_1 \right) = \underline{\frac{3}{2} nR \left(\frac{P_2 V_1}{nR} - T_1 \right)} \end{aligned}$$

4

$$\begin{aligned} \Delta H &= C_P \Delta T = C_P \left(\frac{P_2 V_1}{nR} - T_1 \right) \\ &= \underline{\frac{5}{2} nR \left(\frac{P_2 V_1}{nR} - T_1 \right)} \end{aligned}$$

11)



$P_f = 1 \text{ bar}$

a. A l'équilibre $P_f = P_{\text{ext}}$.

$$\Delta U = W + \cancel{Q} = C_V (T_F - T_I)$$

$$W = -P_{\text{ext}} (V_F - V_I) = -P_F (V_F - V_I)$$

2

$$\Rightarrow C_V (T_F - T_I) = -P_F (V_F - V_I)$$

$$T_F = -\frac{P_F}{C_V} (V_F - V_I) + T_I$$

$$C_V(T_F - T_i) = -P_F(V_F - V_i)$$

$$= -P_F \left(\frac{nRT_F}{P_F} - \frac{nRT_i}{P_i} \right)$$

$$P_i V_i = nRT_i$$

$$P_F V_F = nRT_F$$

$$\frac{3}{2}nR(T_F - T_i) = nR \left(T_i \frac{P_F}{P_i} - T_F \right)$$

$$V_i = \frac{nRT_i}{P_i}$$

$$\frac{3}{2}T_F - \frac{3}{2}T_i = \frac{P_F}{P_i} T_i - T_F$$

$$V_F = \frac{nRT_F}{P_F}$$

$$\left(\frac{3}{2} + 1 \right) T_F = \left(\frac{P_F}{P_i} + \frac{3}{2} \right) T_i$$

$$T_F = \frac{\frac{P_F}{P_i} + 3/2}{5/2} T_i = \frac{3 + \frac{2P_F}{P_i}}{5} T_i$$

AN: $\frac{P_F}{P_i} = 10.$

$$T_F = \frac{10 + 1,5}{2,5} \cdot 573 = \underline{2,6 \cdot 10^3 \text{ K}}$$

(b) $W = C_V(T_F - T_i)$

2

$$= \frac{3}{2}nR \left(1 + \left(\frac{P_F}{P_i} + \frac{3}{2} \right) \frac{2}{5} \right) T_i$$

AN:

$$W = 1,5 \times 8,31 \times \left(1 - \frac{11,5}{2,5} \right) \cdot 573 = \underline{26 \text{ kJ}}$$